CHAPTER 5

DESINGING A HYBRID FRAMEWORK FOR GLAUCOMA SCREENING USING FUNDUS IMAGES

5.1 OVERVIEW OF PROPOSED RESEARCH

Glaucoma screening is challenging since symptoms can not show up until the condition is advanced. A non-invasive technique that works well for mass screening is digital fundus photography (DFP) glaucoma screening. The presence or absence of glaucoma symptoms can be detected by automated systems. OD and optic circle region are isolated for resulting optic circle boundary assessment to test for glaucoma in light of DFP and optic plate qualities. The retinal nerve filaments converge into the optic circle to create a region is recognized as the OC, and the optic plate itself shows up as a splendid, oval or roundabout locale that is somewhat concealed by manners. The cup-to-nipple ratio can be computed and utilized for glaucoma assessment once the OD and OC have been separated. An automated classification method that learns features from DFP-labeled photos can be used to diagnose glaucoma. Adding feature appearance geometry seems to boost outcomes, although performance deteriorates with low-resolution images.

5.2 Problem Credentials

Scientists are trying to improve DL techniques in order to eliminate glaucoma-related vision loss in comparison to conventional state-of-the-art techniques. Each method has a number of shortcomings, like:

- The majority of feature extractions rely on conventional techniques.
- The OD is tough to locate.
- Using a repeating classifier

- When implementing deep frame work, image quality is a major issue.

The aforementioned research gaps are the focus of related publications in the feature extraction and artificial intelligence literature, which also seek to suggest novel methods. In order to overcome the aforementioned downsides we develop a brand-new glaucoma prediction technique that leverages the deep architecture of AAA to enhance performance without being constrained by the conventional way that utilized.

5.3. Proposed Methodology

The following procedures are used to apply the suggested framework in this actual experimental study:

- Segmentation, boundary detection, and Ternausnet OD fitting
- Automatic Gradient Algorithm Optimizer FRCNN (AFRCNN) and FRCNN together
- Use SVM in place of the SoftMax layer
- Use Friedman statistical analysis FRCNN in the AAASVM optimizer for glaucoma screening. The research process work breakdown is presented in Figure 5.1.

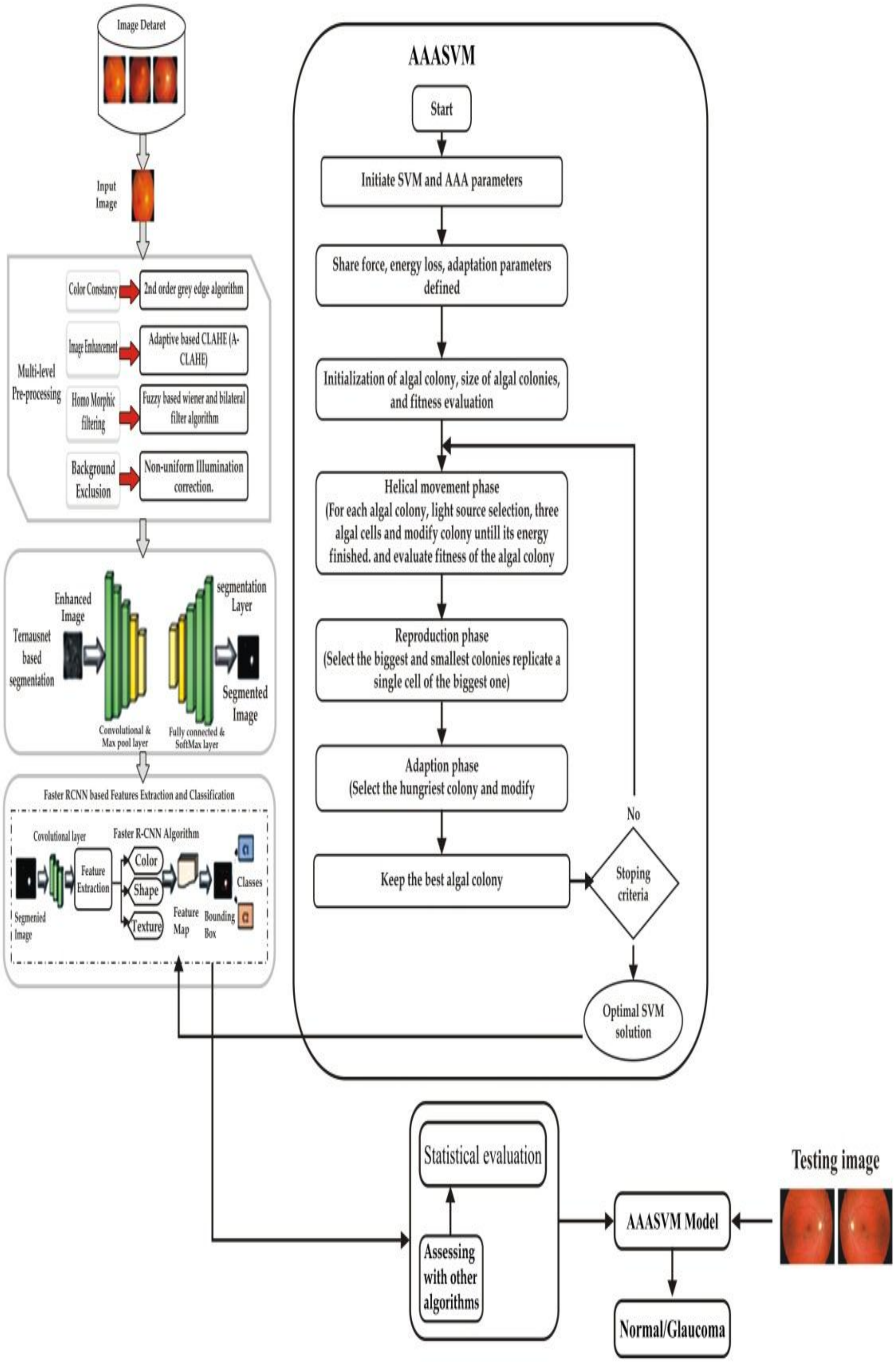

Figure 5.1 Architecture Diagram for AAASVM

5.3.1 Joint Segmentation and Border Detection with Ternausnet OD

To detect OD borders transparently, we present Ternausnet used a VGG11 pretrained encoder model. Seven convolutional layers make up the model, which is arranged according to a modified linear activation function. The feature map produced by these two procedures can be reduced by up to five operations in the pooling layer. The first convolutional layer has a 3 × 3 mask size for 64 channels and 5 × 5 × 3 × 32 weights. The linked set is used to deepen the model, doubling the number of channels to 512. The input picture blocks are then reduced to 1 × 1 convolutions by the encoder after processing 3 × 3 convolutions.

The result is then sent to a decoder, which decreases the number of channels from 512 to 64 and doubles the feature size. The position of OD is determined by the pieces OC and OD.

5.3.2 FRCNN-Based Feature Extraction

For image-based retrieval covers a variety of feature extraction techniques that are helpful for the recently developed FRCNN-based feature extraction. At this point, Ternausnet's picture output is sent into FRCNN in order to extract features like texture, color, and form. Instead of doing a selective search, FRCNN employs RPN. Conv-Net receives the input image and returns anchor boxes of various sizes, feature map values, and object data. A pooling layer known as ROI receives the previous suggestions and reduces their size until it matches the size of the bounding box. The softmax layer receives the prior output following this processing. A sliding window of the extracted characteristics is produced by RPN. Stochastic gradient descent is the method's solution. Faster grouping is achieved by supporting feature vector gradients in the proper direction. This provides the momentum slope's average value.

$$V_t = \beta V_{t-1} + \alpha V_w L(W, X, y) \tag{5.1}$$

and $W = W - V_t$

Here, V is slopes average value, W is weight and β is a hyperparameter with a value between 0 and 1.

In this case, the feature extraction layer is the classification layer. Following, aforementioned procedures, the box regression layer render a box in the feature picture. Using texture characteristics like variance, correlation, energy, and uniform entropy that were taken from the previously mentioned training retina dataset based on FRCNN, it facilitates gray scale intensification. To find abnormalities in the retinal image, color features including mean, variance, standard deviation, slope and kurtosis are then retrieved. After that, shape features are extracted. In order to produce feature maps for predictive glaucoma classification, these features are archived for later analysis.

5.3.3 Glaucoma Screening Using a Modified RCNN and Tuned AFRCNN

To predict glaucoma and normal and abnormal retinal pictures, the extracted feature maps are fed into the FRCNN's classification layer. For efficient classification, fully integrated SoftMax and classification layers are added. The input image's boxes that have been binned by mean pooling are returned by a box regression layer. After completing the aforementioned procedures, the classification layer uses the retinal image's CDRs to predict glaucoma. Utilizing an adaptive gradient technique to increase classification accuracy and do away with manual learning rate modifications. Stochastic gradients are the subject of this kind of optimization, which fixes features that show up more frequently than features that show up infrequently in previous observations. We use each parameter t at each time point of the final slope θ_i to adjust the learning rate in line with the learning rate in equation (5.2)

$$\theta_{t+1,i} = \theta_{t,i} - \frac{n}{\sqrt{G_{t,ii}+\epsilon}} g_{t,i} \tag{5.2}$$

In this case, the squared gradient decay factor is 0.99, the initial learning rate and the maximum epochs trailing on batch size.

5.3.4 SVM

To enhance anchor box performance, SVM is utilized in this work in place of the Softmax classifier in RPN. SVM also has a more profound theoretical underpinning. With fewer training samples, SVM can efficiently identify the global best solution. SVM techniques are frequently applied to pattern recognition, non-linear regression, object detection, and object classification. In order to capture some mapping of nonlinear vectors based on multidimensional spaces, linear models are frequently employed to construct nonlinear class borders. In the multidimensional space, build a suitable separation hyperplane. Consequently, SVM's ability to find the best hyperplane to divide the output classes is one of its advantages. The distinction between foreground and background bins is made using a radial basis function (RBF) kernel in an SVM classifier. There are two constraints in the SVM RBF kernel. Namely, Softmax and C function (parameter-free). Consequently, the SVM method performs better than the Softmax technique in terms of optimisation. Equation (5.3) employs x_i and x_j as kernel input samples, γ as positive scalar parameter controls the width of the kernel and K as a kernel parameter.

$$K(x_i, x_j) = exp\left(-\gamma \left\| x_i - x_j \right\|^2\right) \tag{5.3}$$

5.3.5 AAA

This method was inspired by microalgae activity. The amount of light that algae absorb in order to gather nutrients determines how much they grow. Algal cell clusters that form aggregate populations give rise to colonies. The algal population is Y_i = [$Y_{(i,1)}$,$Y_{(i,2)}$,.. $Y_{(i,D)}$] shows in equation (5.4), and all algal cells in the j^{th} dimension of the i^{th} algal group are recorded as $y_{(\text{i,j})}$.

$$Pop(Population) = \begin{bmatrix} y_{1,1} & \cdots & Y_{1,D} \\ . & \cdots & . \\ . & \cdots & . \\ . & \cdots & . \\ Y_{NP,1} & \cdots & Y_{NP,D} \end{bmatrix} Y_{1,1_{y_{1,1}}} \tag{5.4}$$

In these, *NP* stands for the total number of algae in the population, and *D* for size of the algal colony. Algae move and swarm to areas that are rich in nutrients. Algal colonies try to get better through adaptation, evolution, and mobility. Establishing the colony in the ideal site yields the best result.

Three phases comprise the process: spiraling, adaption, and regeneration or evolution. Algal colonies expand and evolve during the reproductive phase in order to identify appropriate solutions. Algal colonies cannot expand and will eventually perish if they behave without the desired outcomes. The evolutionary adaptations of algal colonies are used to rate them (*G*). The smallest one-dimensional algae are approximately destroyed and exhibit the same dimensions as the largest algae (Figure 5.2).

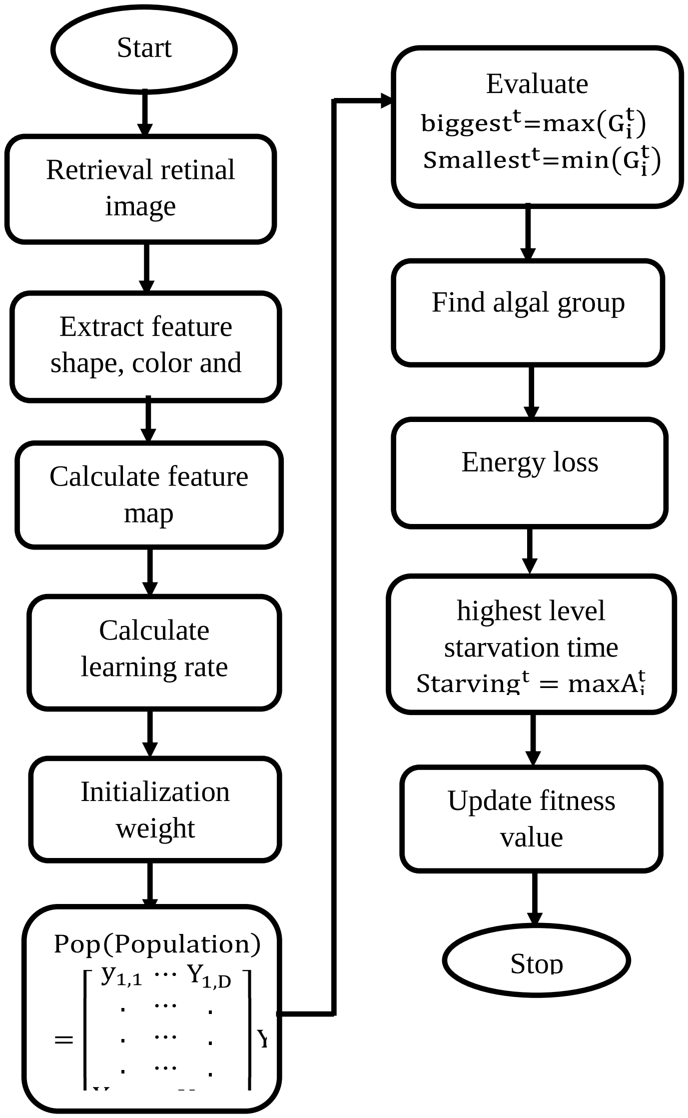

Figure 5.2 AAA Flowchart Diagram

5.3.6 Description of Experimental Datasets

Mat-lab 2020a (The Math Works, Inc., Natick, MA, USA) with 8 GB of RAM is used to accomplish the suggested method. The G1020, high resolution fundus (HRF), and digital retinal image vessel extraction (DRIVE) retinal datasets are used to test the suggested OD boundary detection, OD and OC segmentation, and glaucoma classification.

A. G1020 Retinal Dataset

There are 1,020 retinal pictures of 432 people in this dataset. Out of these, 724 photographs from 322 patients revealed a normal retina, while 296 photos from 110 patients showed glaucoma. This dataset has no practical imaging constraints when compared to other retinal fundus imaging datasets. A set of fundus photos with a 45-degree field of view following an extended fall can be found in this repository.

B. HRF Image Set

This image dataset comprises of 45 retinal fundus photos taken by solid patients with glaucoma and diabetic retina. The pictures were captured with a Group CR-1 fundus camera (Tokyo, Japan) that has a 3304 2336 size and a 45-degree field of vision.

C. DRIVE Retinal Dataset

This dataset contains 400 retinal pictures of diabetes patients between the ages of 25 and 90. Using a Canon CR5 non-mydriatic 3CCD camera, these snaps were taken at a 45-degree angle and stored in JPEG format.

5.3.7 System Workflow

The current research on the predictive categorization of glaucoma involves four steps. To get a better image output, the G1020 image first goes through a multi-stage pre-processing process. The Ternausnet input layer receives preprocessed photos in order to precisely identify OD bounds. For more accurate ROI segmentation, this method uses a $1 \times 1 \times 96$ segmentation layer with encoder, decoder, maxpool, deep concatenation, softmax $227 \times 227 \times 96$, and weights $5 \times 5 \times 32 \times 32$. It is not the same as division-level division. This indicates that OC is the center less the boundary OD. The disc circle's radius and center can then be determined by combining transformations. Each segmented is brought back to the OC and OD bounds after going through the post-processing block, utilizing the vertical distance between OC and OD, CDR is defined. The binary task classification result (represented as I) inferred from the cross-entropy is used to lower the retinal image segmentation, which is compressed using a loss function. From here on, the Jacquard index indication defines the loss function.

The system can optimize the likelihood of correctly forecasting the pixel value in the cross-sectional area and minimize the loss function. In this case, the feature map is fed into the RPN, which yields a score for the bounding box probability at the disc boundary as well as some candidate locations. The input image and each pixel value that matches the region of interest's detection probability determine the output image's prediction. The threshold is set at 0.3 in accordance with the loss function. Based on 255 multiples per pixel, the expected OD boundary image is currently shown in black and white. The threshold is less than the supplied value if the pixel value is zero, and the pixel is larger than the defined threshold if the pixel value is greater.

To retrieve the objects (OC and OD) in order, the suggested method uses FRCNN to collect color, shape, and texture information. RPN uses feature maps of shape, color, and texture to indicate potential regions. The chance that the square lies on OD and OC is shown by the score assigned to each hypothesis. Use a sliding window to create anchor points, and build anchor points to form a bounding box. The convolution's coded coordinates, not the bounding box's coded coordinates, are the output. Our objective is to use regional recommendations based on form, color, and texture scores to infer retinal images (OD and OC). The classifier's emphasis on the feature map's bounding boxes, pooling areas, and maximum pooling of $m \times m$ region size is checked using this ROI.

A classifier in the deep convolutional layer uses the bounding box coordinates of the produced encoding to indicate the likelihood of object class. Here, score probabilities are generated using the Softmax layer. The correlation between the RPN candidate bounding box coordinates and its coded coordinates is used to determine the best bounding box prediction. The system evaluates the performance of glaucoma classification by calculating the ROC curve and AUROC for a particular dataset after differentiating between normal and glaucomatous retinal pictures. A new design architecture is said to perform well if its AUROC is high.

5.3.8 Evaluation of OD Boundary Detection and Joint Segmentation Results

To increase the suggested segmentation's dependability, performance measures are added. The background region is initialised to True Negatives (TN) and False Negatives (FN), while the output of the segmented material is initialised to True Positives (TP) and False Positives (FP) for OD/OC boundary. A confusion matrix is created to evaluate performance.

Figure 5.3 displays Ternausnet's segment output. This approach is contrasted with the FRCNN method, Unet-based segmentation, superpixel method, and quadratic differential regularization (SVM).

The segmentation task is organized as bounding box detection, and a comparison chart is created based on the OD and OC errors. This strategy makes complicated techniques quick and simple. Transfer learning can be used with or without pre-trained models in this approach. This task reduces mistakes in the OD and OC segment's proper disk boundaries (Table 5.1). The glaucoma trial in this study comprised a number of performance metrics. The performance requirements are presented in Equations (5.5) through (5.10).

$$A_{ccuracy} = \frac{(T_P+T_N)}{(T_P+F_P+T_N+F_N)} \tag{5.5}$$

$$S_{ensitivity} = \frac{(T_P)}{(T_P+T_N)} \tag{5.6}$$

$$s_{pecificity} = \frac{(T_N)}{(T_N+T_P)} \tag{5.7}$$

$$P_{recision} = \frac{(T_p)}{(T_P+F_P)} \tag{5.8}$$

$$R_{ecall} = \frac{(T_P)}{(T_P+F_N)} \tag{5.9}$$

$$F1_{Score} = \frac{2\times Pre\times Rec}{Pre\times Rec} \tag{5.10}$$

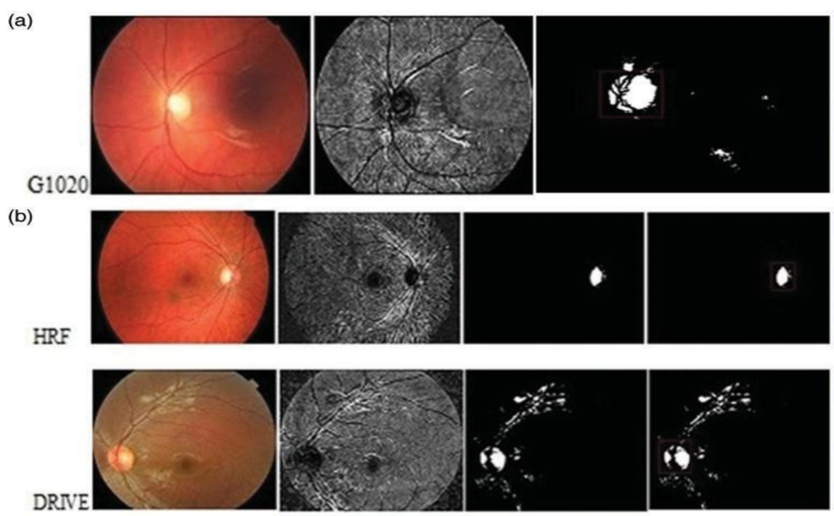

Figure 5.3 (a), (b) Ternausnet + Frcnn Glaucoma Screening of Three Retinal Dataset

Table 5.1 Assessment of OD and OC Segmentation Using Various Methods

Method	Disc Error	Cup Error
Superpixel	0.102	0.264
Quadratic divergence SVM	0.11	–
Unet	0.115	0.287
FRCNN	0.069	0.222
Proposed TernausNet	0.061	0.207

5.4 RESULTS AND DISCUSSION

The FRCNN classifier's performance on the learning set (G1020) and examination set (HRF and DRIVE), the newly suggested framework's accuracies are 95.11%, 93.7%, and 92.87%, respectively. As shown in Figure 5.4, the G1020 has an AUROC of 0.92, HRF of 0.84, and a classification of 0.87 for the Drive Retina dataset. By lowering the Mean Square Error (MSE), aforementioned output is sent towards an optimizer (AAA) to increase the glaucoma classifier's accuracy.

The estimation accuracy is increased by using AAA as a solution for SVM-FRCNN (Table 5.2). It improves accuracy and speeds up learning. For each dataset, this optimizer increased accuracy by 1.02%. For all training and test sets, the AUROC ranges from 0.87 to 0.93 (left). Compared to standard methods, this cuts the estimation time by 0.02 seconds. The performance evaluation of FRCNN, optimal AFRCNN, SVM, and AAASVM.The accuracy of the AAAASVM approach is higher than that of FRCNN, untuned FRCNN, and SVM.

5.4.1 Performance Evaluation Based on the Friedman Test

For evaluation, a null hypothesis was formulated and the classifiers' AUROC probability was compared to 0.05[-47]. Since all procedures are equal and of the same rank, this is known as the null hypothesis. In this case, the classifier with the highest AUROC value is ranked first and is referred to as the best classifier.

$$R_i = \frac{1}{N}\sum_{j-1}^{N} r_j^i \quad (5.11)$$

where r_j^i is the number of classifiers, N is the number of datasets utilised, and k = 4 is the j^{th} dataset of the i^{th} rank of the classifier. The resultant value for chance of 0.05 was greater than the chi-square value.

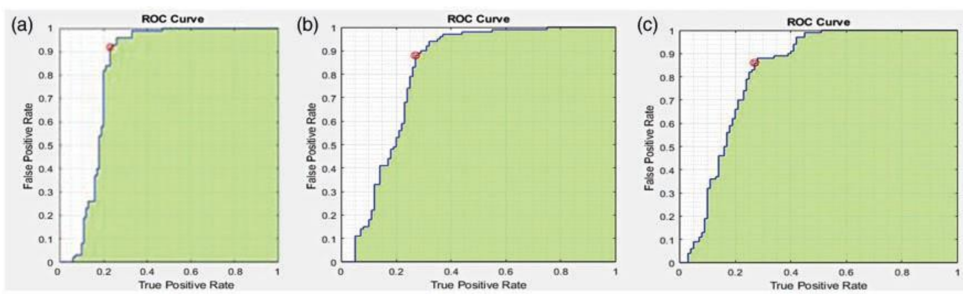

Figure 5.4 Result of AAA-SVM -AUC on ROC of (a) G1020 (b) HRF (c) DRIVE

Consequently, the degree to which each classifier rejected the null hypothesis determined its ranking. This suggests that AAASVM outclassed the other classifiers in standings of AUROC. Currently recommended method can be a helpful tool for assessing the evolution of glaucoma classification at the primary stage, improving glaucoma prediction, with a higher ideal accuracy of 96.55 (Figure 5.5).

Table 5.2 Different Classifiers Performance

Classifiers	Average Sensitivity (%)	Average Specificity (%)	Average Accuracy (%)
FRCNN	93.415	92.112	94.88
AFRCNN	94.352	93.062	95.87
SVM	94.050	95.022	95.09
AAASVM	93.855	95.707	96.55

AAASVM, or artificial algae algorithm with support vector machine; AFRCNN, or adaptive gradient algorithm optimiser, is a faster region-based convolutional neural network.

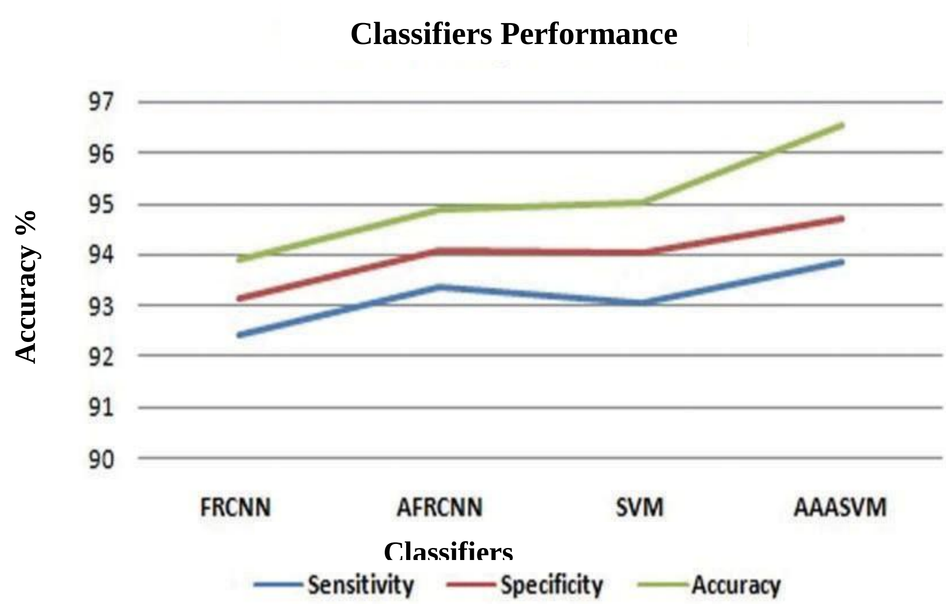

Figure 5.5 Performance Metrics of the Different Classifiers

A quicker region-based convolutional neural network is called AFRCNN, or adaptive gradient algorithm optimizer, AAASVM stands for artificial algae algorithm with SVM. Table 5.3 lists some previous classifiers for comparison.

Table 5.3 Assessment in Comparison with Other Glaucoma Screenings

Authors	**Classifiers**	**No. of Features**	**Accuracy (%)**
Koh et al.[88]	RF	15	92.48
Mookiah et al. [89]	SVM	35	95
Kasu et al. [90]	MLP	4	97.6
Dua et al. [91]	SMO	14	93.3
Elangovan and Nath [92]	CNN		96.6
Proposed	Faster rcnn, adafrcnn AAASVM	14	96.15 **97.01**

5.5 SUMMARY

Ternausnet is utilized in this study for combined OD and OC segmentation and effective OD border detection. By diagnosing OD, which is quite challenging, we often get very low error when predicting the onset of glaucoma using three retinal datasets. It reduces inference time and becomes an efficient computer decision support system by integrating fundus pictures with computer vision techniques, which are improved by AAASVM. One benefit of this system is that, like conventional approaches, it may be implemented with any processor and does not require DL.